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CASE STUDY:2

TOPIC: URBAN AIR QUALITY IN INDIAN CITIES – EXPLORATORY DATA ANALYSIS AND HYPOTHESIS TESTING

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DATA SCIENCE

RAGAVI M

Reg. No: 126252010

M.Tech CSE

Submitted to,

Ms.Dr.M.PRABHA

Associate Professor CSE

Urban Air Quality in Indian Cities – Exploratory Data Analysis and Hypothesis Testing

Domain: Environmental Data Analytics

SDG 11: Sustainable Cities and Communities

SDG 13: Climate Action

Dataset: CPCB Urban Air Quality Data (2015–2020)

Tools: Python, Pandas, SciPy, Statsmodels, Seaborn

Project Type: Exploratory Data Analysis and Statistical Hypothesis Testing

URBAN AIR QUALITY IN INDIAN CITIES – EXPLORATORY DATA ANALYSIS AND HYPOTHESIS TESTING

1. ABSTRACT

Air pollution is one of the major environmental challenges affecting urban areas in India. Increasing industrial activities, vehicle emissions, construction, and population growth contribute to variations in air pollutant concentrations. Monitoring air quality is essential for understanding environmental conditions and supporting sustainable urban development.

This project focuses on Exploratory Data Analysis (EDA) and statistical hypothesis testing using publicly available Central Pollution Control Board (CPCB) air quality data collected from Indian cities between 2015 and 2020.

The main objective is to examine the distribution and variation of major air pollutants, particularly Particulate Matter 2.5 (PM2.5) and Nitrogen Dioxide (NO_2), across different cities and seasons.

The project uses Python libraries such as Pandas, SciPy, Statsmodels, and Seaborn to perform data preprocessing, statistical summarization, visualization, confidence interval estimation, and hypothesis testing.

A comparative statistical analysis will be performed between the corresponding lockdown period in 2019 and 2020 to investigate whether the COVID-19 lockdown period was associated with a significant change in pollutant concentrations.

Additionally, correlation analysis will be conducted to understand relationships between different air pollutants. The project will also explain why correlation does not necessarily indicate causation.

The expected outcome is a comprehensive understanding of urban air quality patterns, seasonal variations, pollutant relationships, and statistical differences during the lockdown period.

Keywords: Air Quality, Exploratory Data Analysis, PM2.5, NO_2, Hypothesis Testing, Confidence Interval, Correlation Analysis, COVID-19 Lockdown, Environmental Analytics.

2. INTRODUCTION

Air pollution is a serious environmental issue affecting human health, ecosystems, and the quality of life in urban areas. Indian cities experience different levels of air pollution due to transportation, industrial emissions, construction activities, and changing weather conditions.

Air quality monitoring helps identify pollution levels and understand how environmental conditions change over time.

The Central Pollution Control Board (CPCB) monitors air quality through monitoring stations located across different Indian cities. The collected information includes measurements of several pollutants such as PM2.5, PM10, NO_2, SO_2, CO, and O_3.

However, simply collecting air quality data is not sufficient. The data must be analyzed to identify patterns, variations, relationships, and significant differences.

Exploratory Data Analysis (EDA) provides an effective approach for understanding datasets through statistical summaries and visualizations.

Statistical hypothesis testing helps determine whether observed differences between two groups are statistically significant or could have occurred due to random variation.

This project uses air quality measurements from 2015 to 2020 to study pollution distributions, seasonal variations, pollutant correlations, and changes during the COVID-19 lockdown period.

2.1 Domain

Environmental Data Analytics.

2.2 Sustainable Development Goals

SDG 11 – Sustainable Cities and Communities

The project supports sustainable urban development by examining air pollution patterns in Indian cities.

SDG 13 – Climate Action

The project contributes to environmental awareness by analyzing changes in air pollutant concentrations and discussing the importance of monitoring environmental conditions.

3. PROBLEM STATEMENT

Urban air pollution levels vary across Indian cities and seasons due to differences in traffic, industrial activities, weather conditions, and other environmental factors.

Although air quality monitoring produces large amounts of data, identifying meaningful patterns and determining whether observed changes are statistically significant require proper data analysis techniques.

The major problems addressed in this project include:

1. Understanding the distribution of PM2.5 and NO_2 across Indian cities.
2. Identifying seasonal variations in air pollutant concentrations.
3. Measuring statistical uncertainty using confidence intervals.
4. Determining whether pollutant concentrations during the 2020 COVID-19 lockdown differed significantly from the corresponding period in 2019.
5. Identifying relationships between different air pollutants through correlation analysis.
6. Understanding the limitations of correlation-based conclusions.

Therefore, the problem addressed by this project is to analyze urban air quality patterns across Indian cities using exploratory data analysis and statistical techniques, and to investigate whether the 2020 lockdown period was associated with statistically significant changes in pollutant concentrations compared with the corresponding period in 2019.

4. OBJECTIVES

The major objectives of this project are:

1. To collect and examine public air quality data from Indian cities between 2015 and 2020.
2. To perform exploratory data analysis on PM2.5 and NO_2 concentrations.
3. To summarize pollutant distributions across different cities.
4. To analyze seasonal variations in air pollution levels.
5. To calculate confidence intervals for selected pollutant measurements.
6. To compare pollutant concentrations during the 2020 lockdown period with the corresponding period in 2019.
7. To perform independent t-tests or Mann–Whitney U tests based on the data distribution and sampling design.
8. To examine relationships between different air pollutants using correlation analysis.
9. To interpret statistical findings without assuming that correlation implies causation.

10. To present the findings through visualizations and a statistical test summary.

5. DATASET DESCRIPTION

5.1 Dataset Details

The project uses publicly available urban air quality measurements collected from CPCB monitoring stations across Indian cities.

A compiled dataset covering the period 2015–2020 is available through the Transformative Urban Mobility Initiative (TUMI) Data Portal.

Attribute	Description
Dataset name	Urban Air Pollution Measurements, India
Original data provider	Central Pollution Control Board (CPCB)
Dataset source	TUMI Data Portal
Domain	Environment
Geographic coverage	Multiple Indian cities
Time period	2015–2020
Data frequency	Hourly and daily
Data format	CSV
Main purpose	Urban air quality analysis

The dataset contains measurements of multiple air pollutants collected from monitoring stations across Indian cities.

5.2 Important Attributes

The following attributes will be considered for the analysis:

1. Location Information

- City
- Monitoring station

2. Time Information

- Date
- Year
- Month
- Season
- Day

3. Air Pollutant Information

- PM2.5
- PM10
- NO_2
- NO
- NOx
- SO_2
- CO
- O_3

4. Air Quality Information

- AQI
- AQI category, where available

5.3 Major Pollutants

PM2.5 (Particulate Matter 2.5)

PM2.5 refers to fine airborne particles with a diameter of 2.5 micrometres or smaller. These particles can penetrate deep into the respiratory system.

NO_2 (Nitrogen Dioxide)

Nitrogen dioxide is a gaseous air pollutant mainly associated with fuel combustion, vehicle emissions, and industrial activities.

These two pollutants will be the primary focus of the project.

6. PROPOSED METHODOLOGY

The project follows a systematic statistical analysis workflow.

6.1 Overall Workflow

Public CPCB Air Quality Dataset
↓
Data Collection and Loading
↓
Data Inspection and Preprocessing
↓
Exploratory Data Analysis
↓
City-Wise Pollutant Distribution
↓

Seasonal Variation Analysis
↓
Confidence Interval Calculation
↓
2019 vs 2020 Lockdown Comparison
↓
Hypothesis Testing
↓
Pollutant Correlation Analysis
↓
Statistical Interpretation and Reporting

6.2 Data Collection and Loading

The public air quality dataset will be downloaded and imported into Python using Pandas.

The initial analysis will include:

- Number of records and columns.
- Available Indian cities.
- Date range.
- Pollutant attributes.
- Data types.
- Missing values.
- Duplicate records.

6.3 Data Preprocessing

Before performing statistical analysis, the dataset will be examined and prepared.

The following operations will be performed:

1. Handle missing pollutant measurements.
2. Remove or investigate duplicate observations.
3. Convert date columns into appropriate datetime formats.
4. Identify invalid or negative pollutant measurements.
5. Standardize city names.
6. Separate year and month information.
7. Create seasonal categories.

The cleaning process will preserve genuine environmental observations and document any exclusions.

6.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is used to understand the structure, distribution, and characteristics of the dataset.

The following statistical measures will be calculated:

- Mean.
- Median.
- Minimum.
- Maximum.
- Standard deviation.
- Quartiles.
- Interquartile range.

Visualizations will include:

- Histograms to understand pollutant distributions.
- Boxplots to identify variations and extreme observations.
- Bar charts to compare city-wise average pollutant levels.
- Line charts to examine yearly trends.
- Heatmaps to visualize pollutant relationships.

6.5 City-Wise Pollutant Distribution

This analysis will summarize PM2.5 and NO_2 concentrations across different Indian cities.

The mean and median pollutant concentrations will be calculated for each city.

The results will help identify:

- Cities with relatively higher PM2.5 levels.
- Cities with relatively higher NO_2 levels.
- Differences in pollution distributions.
- Variation in pollutant concentrations between cities.

Bar charts and boxplots will be used to compare city-wise pollution levels.

6.6 Seasonal Variation Analysis

Air pollution levels can change across seasons due to variations in temperature, wind, rainfall, and human activities.

The dataset will be grouped into four seasons:

Season	Months
Winter	December, January, February
Summer	March, April, May

Monsoon	June, July, August, September
Post-monsoon	October, November

Average PM2.5 and NO_2 concentrations will be calculated for each season.

Seasonal comparisons will help identify periods with relatively higher or lower pollution levels.

A boxplot or seasonal bar chart will be used to present the results.

6.7 Confidence Interval Calculation

A confidence interval provides a range of plausible values for an unknown population parameter, such as the mean pollutant concentration.

A 95% confidence interval will be calculated for selected PM2.5 and NO_2 measurements.

The confidence interval for a mean using the t-distribution is:

$$95\%\ CI = Mean \pm t \times (Standard\ Deviation / \sqrt{n})$$

Where:

- Mean = Sample mean.
- t = Critical t-value.
- Standard Deviation = Sample standard deviation.
- n = Sample size.

Confidence intervals will help understand the uncertainty associated with estimated average pollution levels.

6.8 Lockdown Period Comparison (2019 vs 2020)

The COVID-19 pandemic resulted in restrictions on transportation, industrial activities, and public movement across India.

The project will investigate whether air pollutant concentrations changed during the lockdown period in 2020 compared with the corresponding dates in 2019.

The comparison will focus on:

- PM2.5 concentrations.
- NO_2 concentrations.

For a consistent comparison, the corresponding calendar period will be selected for both years.

For example:

- 2019: March 25 to May 31.
- 2020: March 25 to May 31.

The average pollutant concentrations will be calculated for both periods.

The results will be visualized using comparative boxplots and bar charts.

6.9 Hypothesis Testing

Hypothesis testing will be used to determine whether the differences in pollutant concentrations between the two periods are statistically significant.

Two statistical tests will be considered.

A. Independent Samples t-test

This test compares the means of two independent groups.

It will be considered when the assumptions of the test are reasonably satisfied.

B. Mann–Whitney U Test

This is a non-parametric test used to compare two independent groups when a distribution-free alternative is appropriate.

The selection of the test will depend on the distribution, sample structure, and independence of observations.

Hypotheses:

Null Hypothesis (H_0):

There is no statistically significant difference in PM2.5 and NO_2 concentrations between the corresponding 2019 and 2020 lockdown-period observations.

Alternative Hypothesis (H_1):

There is a statistically significant difference in PM2.5 and NO_2 concentrations between the corresponding 2019 and 2020 lockdown-period observations.

Significance Level:

$\alpha = 0.05$

Decision Rule:

- If p-value < 0.05, reject the null hypothesis.

- If p-value ≥ 0.05, fail to reject the null hypothesis.

The results will be interpreted separately for PM2.5 and NO_2.

Where possible, the analysis will compare matched city-monitoring stations and corresponding calendar dates. This helps reduce differences caused by location and season.

6.10 Correlation Analysis

Correlation analysis is used to examine the strength and direction of relationships between numerical variables.

The project will study relationships between:

- PM2.5 and PM10.
- PM2.5 and NO_2.
- NO_2 and NO.
- Other available pollutant combinations.

Pearson correlation will be used to examine linear relationships. Spearman correlation may also be considered for monotonic relationships or data with strong non-normality and outliers.

Correlation coefficients range from -1 to +1.

Correlation	Interpretation
+1	Perfect positive correlation
0	No linear correlation
-1	Perfect negative correlation

A correlation heatmap will be generated using Seaborn.

Why Correlation Does Not Mean Causation

A strong relationship between two pollutants does not prove that one pollutant causes the other.

For example, PM2.5 and NO_2 may increase together because both are influenced by common emission sources such as traffic.

Other factors, including temperature, wind speed, rainfall, and industrial activity, may also influence pollutant concentrations.

Therefore, correlation analysis identifies relationships but cannot independently establish cause-and-effect relationships.

6.11 Statistical Interpretation

The final stage will interpret all the statistical findings.

The interpretation will include:

- City-wise pollution differences.
- Seasonal pollution patterns.
- Estimated mean pollutant concentrations.
- Confidence intervals.
- Hypothesis test results.
- Correlation coefficients.
- Environmental explanations and limitations.

The findings will be summarized in a statistical test summary table.

7. TOOLS AND TECHNOLOGIES

The following tools and libraries will be used:

Tool	Purpose
Python	Main programming language
Pandas	Data loading, cleaning and grouping
SciPy	Statistical tests and statistical calculations
Statsmodels	Confidence intervals and statistical analysis
Seaborn	Statistical data visualization
Matplotlib	Charts and graphs
Jupyter Notebook	Interactive analysis and documentation

7.1 Software Requirements

- Python 3.x
- Jupyter Notebook
- Pandas
- SciPy
- Statsmodels
- Seaborn
- Matplotlib

8. EXPECTED OUTCOMES

The project is expected to produce the following results:

1. A clear understanding of PM2.5 and NO_2 distributions across Indian cities.
2. Identification of seasonal variations in air pollution.
3. City-wise statistical summaries of pollutant concentrations.

4. Calculation of 95% confidence intervals.
5. Statistical comparison of the corresponding 2019 and 2020 lockdown periods.
6. Identification of significant differences using appropriate hypothesis tests.
7. Visualization of relationships between different air pollutants.
8. Interpretation of correlation without assuming causation.
9. A comprehensive understanding of urban air quality patterns between 2015 and 2020.

9. DELIVERABLES

The project will produce three major deliverables.

9.1 EDA Notebook

A Jupyter Notebook containing:

- Dataset loading.
- Data preprocessing.
- Statistical summaries.
- City-wise analysis.
- Seasonal analysis.
- Confidence interval calculations.
- Lockdown-period comparison.
- Hypothesis testing.
- Correlation analysis.
- Visualizations.

9.2 Statistical Test Summary

A summary containing:

- Pollutant name.
- Comparison period.
- Statistical test used.
- Test statistic.
- p-value.
- Significance decision.
- Interpretation.

9.3 Five-Minute Project Briefing

A short presentation explaining:

1. Project introduction and objectives.
2. Dataset and major pollutants.

3. EDA and seasonal findings.
4. Confidence intervals and hypothesis testing.
5. Correlation analysis and conclusion.

10. CONCLUSION

This project proposes an exploratory and statistical analysis of urban air quality across Indian cities using publicly available CPCB air quality data from 2015 to 2020.

The study focuses on understanding the distribution of PM2.5 and NO_2, identifying seasonal variations, calculating confidence intervals, and comparing pollutant concentrations during the corresponding 2019 and 2020 lockdown periods.

Statistical tests such as the independent samples t-test and Mann–Whitney U test will be used to evaluate whether observed differences are statistically significant.

Correlation analysis will provide additional insights into relationships between air pollutants while highlighting the difference between statistical association and causation.

The project will provide useful insights into environmental data patterns and demonstrate the practical application of exploratory data analysis and statistical hypothesis testing in environmental analytics.

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